

## ACTIVITY 1 FINALS

CODE:

```
1 from LinkedDeque import LinkedDeque as Deque
2 from LinkedQueue import LinkedQueue as Queue
3 from LinkedStack import LinkedStack as Stack
4
5
6 D = Deque()
7 Q = Queue()
8 S = Stack()
9
10
11 D.insert_last(1)
12 D.insert_last(2)
13 D.insert_last(3)
14 D.insert_last(4)
15 D.insert_last(5)
16 D.insert_last(6)
17 D.insert_last(7)
18 D.insert_last(8)
19
20
21 Q.enqueue(D.delete_first())
22 Q.enqueue(D.delete_first())
23 Q.enqueue(D.delete_first())
24
25
26 Q.enqueue(D.delete_first())
27 Q.enqueue(D.delete_first())
28
29
30 while not D.is_empty():
31     Q.enqueue(D.delete_first())
32
33
34 print("Deque D after reordering using Queue:")
35 while not Q.is_empty():
36     D.insert_last(Q.dequeue())
37
38
39 for element in D:
```

```

40     for element in D:
41         print(element)
42
43     D = Deque()
44     D.insert_last(1)
45     D.insert_last(2)
46     D.insert_last(3)
47     D.insert_last(4)
48     D.insert_last(5)
49     D.insert_last(6)
50     D.insert_last(7)
51     D.insert_last(8)
52
53
54     S.push(D.delete_first())
55     S.push(D.delete_first())
56     S.push(D.delete_first())
57
58
59     S.push(D.delete_first())
60     S.push(D.delete_first())
61
62
63     while not D.is_empty():
64         S.push(D.delete_first())
65
66
67     D.insert_first(S.pop())
68     D.insert_first(S.pop())
69
70
71     while not S.is_empty():
72         D.insert_first(S.pop())
73
74
75     print("Deque D after reordering using Stack:")
76     for element in D:
77         print(element)

```

#### OUTPUT:

```

Deque D after reordering using Queue:
1
2
3
5
4
6
7
8
Deque D after reordering using Stack:
1
2
3
5
4
6
7
8

Process finished with exit code 0

```